

[DC]VR Ecology: Relational Haptic Co-Creation for Kinaesthetic Creativity

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ABSTRACT

VR Ecology investigates *relational presence* in XR co-creation: how an AI partner can stay legible and negotiable within embodied making. Instead of prompts or speech, the AI uses a small ProTactile-inspired set of haptic micro-signals (orientation, offer/handoff, slow/confirm). An end-to-end WebXR prototype streams hand and gesture data, drives haptic output, and records trace-memory logs with human/AI/system provenance for replay and annotation. Planned within-subject studies compare visual-only, haptic-only, and combined cues, measuring coordination timing (turn-taking, interruption, response latency), task/process outcomes, and validated self-report of flow and agency; capability-staging contingency matrices keep evaluations comparable as the system evolves.

Index Terms: H.5.2 [Information Systems]: Information Interfaces and Presentation—User Interfaces; I.2.m [Computing Methodologies]: Artificial Intelligence—Miscellaneous; haptics, VR, WebXR, co-creation, creativity support tools, multimodal, accessibility

1 INTRODUCTION AND MOTIVATION

XR—spanning VR/AR, mixed reality—still gets described as a visual medium, as if it were just another screen. But inside the headset, experience arrives as personal somatic spacetime: you lean, you reach, you hesitate, you feel a pull. The craving for the material—for something that pushes back—is far more sensible here than on a desktop. For me, this is not a side note; it is the point. XR becomes a site for studying creativity as movement-based practice, aligned with embodied interaction in HCI [5] and accounts that treat movement and action as foundational to meaning-making [10, 11, 13], including affective dynamics that emerge in and through motion [9]. At the same time, the current wave of generative AI has a habit of pulling creative practice back toward text. Prompting can be useful, but it also asks makers to stop moving, stop listening, and start explaining. My dissertation asks what an AI collaborator looks like when the primary interface is the body: can an AI stay “in the room” with you, share tempo, and still leave you holding the steering wheel? A practical observation underlying this agenda is that AI partners often act not as mere obedient assistants but as non-human collaborators, exhibiting unique timing and presence. They are partially opaque, differently timed, and often “present” through their effects rather than through speech. As Lanier notes, VR has always been about being in a world—not just looking at one [8]. If an AI is going to be a partner inside that world, its intentions need to be available at the level of ongoing action: readable as timing, spatial conduct, and felt consequence.

Scientific framing. I treat *relational presence* as an empirical coordination problem: whether AI intent becomes legible enough to support timely, consensual turn-taking during embodied making. In planned studies, cue modality (visual vs haptic vs combined) and cue type (orientation/offer/slow-confirm) are experimental factors.

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Outcomes include coordination timing (turn-taking, interruption, response latency), task/process measures, and validated self-report of flow and agency. To design for this unique form of interaction, I draw inspiration from touch-first communication models and interspecies coordination, which offer insights into non-verbal interaction. ProTactile communication offers a systematic model of tactile grounding and co-presence through phatic contact and back-channeling [1]. Horse-human and dog-human coordination offer parallel examples of how meaning and trust can be established through subtle movement, contact, and regulation rather than explicit language [6, 7]. In VR Ecology, I do not use these as metaphors to decorate an AI. I treat them as resources (and constraints) for defining what “legible intent” and “shared consent” should look and feel like in XR. My research questions evaluate whether haptic micro-signals improve co-creative coordination and perceived agency/flow compared to visual-only cues, characterize breakdowns (missed/overloaded/misinterpreted cues), and repair behaviors.

2 VR ECOLOGY AS RELATIONAL CO-CREATION

VR Ecology frames the creative studio as an ecology: a shifting field of signals, tools, and reciprocal commitments made in real time. The central hypothesis is that co-creative XR systems improve coordination when AI intent is legible and confirmable in real time, and that haptic micro-signals can provide this legibility with less disruption to embodied flow than visual overlays alone. This aligns with systems that provide (i) rich sensorimotor channels for expression, (ii) legible AI intent (so the human can anticipate and respond), and (iii) mechanisms for maintaining mutual grounding when interaction gets fast and messy. Presence research suggests realistic behaviour can arise when systems support both place illusion and plausibility [12]. Enactive and sensorimotor accounts, meanwhile, emphasize that cognition is not internal representation alone but skilful coupling of body and world [10, 13]. VR Ecology translates these views into a concrete commitment: the AI partner should act as a perceivable participant whose moves are expressed through timing, spatial conduct, and touch.

Design lens: touch-first grounding and calming-signal coordination. Touch-first approaches treat co-presence as something you do: taps, patterned pulses, and brief confirmations can maintain shared context under time pressure [1]. Interspecies interaction offers parallel patterns for orienting attention, regulating arousal, and negotiating turn-taking without words [6, 7]. Accordingly, the haptic vocabulary is organized around three coordination functions: orientation (where to attend), negotiation (offer/handoff/refusal), and regulation (confirm/slow/settle). The cue is an invitation rather than a command: it must be confirmable, ignorable, and repairable in-the-moment.

3 SYSTEM AND METHODOLOGY

The research employs an iterative approach combining research-through-creation and system building, complemented by empirical studies conducted in a single-user creative studio environment. In this loop, the prototype is not “just a demo” but a methodological site of inquiry: I build to think, test to learn, then rebuild. The technical stack is intentionally modular so experimental variations (new cues, mappings, or AI behaviours) can be swapped in without destabilizing runtime performance during immersive sessions.

WebXR Co-Creative Studio The core artifact is a WebXR making environment for gesture-driven 3D mark-making, tool switching, and dynamic scene content. It is implemented with Three.js in the client and a Node/WebSocket runtime for low-latency streaming and coordination. Design choices follow established guidance on 3D UI interaction techniques and evaluation [2], while staying lightweight enough to support frequent build-and-test cycles—the quick step-in, step out, step back in rhythm that is hard to avoid when you are creating from within the world.

Multimodal Sensing and Gesture Fusion The studio integrates live hand tracking (via a native bridge) and additional motion or bio streams when available. A gesture-fusion layer aggregates controller and HMD kinematics, hand joint data, and event density into features used for adaptive visualization (e.g., trails and heatmaps) and for tagging the session log. These features are designed to be interpretable and minimally invasive, supporting later analysis rather than relying on opaque prediction.

Touch-First Haptic Language for Relational Presence A key contribution is a learnable tactile vocabulary that lets the AI partner communicate intent, uncertainty, and invitations to act without relying on text overlays or speech. Building on haptics research on designing distinct touch sensations [4], the system explores patterned pulses, directional sweeps, and intensity modulation mapped to interaction states (e.g., “follow”, “pause”, “hand-off”, “attention here”). In the studies, these tactile patterns function as experimental factors (cue modality and cue type), not as metaphor: they are designed to be discriminable, learnable, and measurable in terms of response timing and downstream action. In effect, it becomes a call-and-response loop: the AI taps, the maker moves, and that movement returns as new context for the next move. The point is not to make the AI louder; it is to make it graspable. Thus the “language” is deliberately small at first. ProTactile emphasizes phatic signaling and back-channeling—confirmations that can be repeated, learned, and repaired when misunderstood [1]. That becomes my baseline: haptics as grounding under time pressure, not as spectacle. I also use interspecies coordination as a heuristic check: can a cue be understood as a calming or orientation signal rather than as an instruction that commandeers the maker’s agency [6, 7]?

Trace-memory Logging and Research Dashboard All events (gesture, tool changes, AI actions, and haptic outputs) are timestamped and stored with source metadata. A post-session dashboard supports filtering, replay, and annotation, enabling the researcher to treat interaction as a time-based conversation rather than a sequence of isolated commands. I think of this as trace-memory: not just what happened, but how initiative shifted, where misunderstandings occurred, and how they were repaired. This is essential for studying mixed-initiative dynamics where creative control shifts fluidly [14].

4 PLANNED STUDIES AND EXPECTED OUTCOMES

Progress to Date: I have implemented an end-to-end prototype that streams hand and gesture data into the WebXR scene, drives wearable haptic output, and records structured logs for post-session review. The current build supports dual-hand gesture recognition, real-time trail visualization modes, and a modular event schema that distinguishes human-, AI-, and system-initiated actions; a small set of tactile intents is in place (orientation, offer/handoff, slow/confirm). Contingency matrices define progressive capability levels for gesture mapping, haptic communication, data capture, and AI-assisted scene behaviour to control complexity during evaluation while preserving internal validity across evolving system versions.

The next phase will conduct a sequence of studies in the studio, beginning with within-subject comparisons of (a) visual-only AI cues, (b) haptic-only cues, and (c) combined cues during short co-creative tasks. Outcomes include task/process measures, coordination timing patterns (turn-taking, interruption, response latency), and validated self-report instruments related to flow and agency [3].

A second phase will focus on haptic legibility and repair: learning curves for a small tactile lexicon (recognition, recall, and in-situ use), and how cues shape repair after misunderstandings. I am particularly interested in whether touch-first back-channeling reduces disruption during creative flow [1] and supports consensual negotiation when the AI proposes a move. Longer-term, I plan to extend the vocabulary by analyzing coordination patterns from touch-first and interspecies contexts and testing whether those patterns translate into stronger grounding, trust, and perceived partnership in XR [6, 7]. Analysis will triangulate condition effects with trace-memory logs to locate initiative shifts and repair.

Expected contributions: (1) a design framework for relational co-creation in XR, emphasizing movement and touch; (2) a modular WebXR research platform for mixed-initiative interaction studies with trace-memory logging; (3) empirical insights into the impact of tactile intent signals and adaptive visual overlays on co-creative coordination; and (4) methodological guidance for integrating practice-based iteration with quantitative and qualitative analysis in immersive creative settings. Not to erase the human touch, but to enliven it. **DC feedback sought:** (1) What study designs best capture grounding, trust/partnership yet maintain embodied flow? (2) What kinds of generalization are credible for longitudinal system-building in XR? (3) How do we validate learnability across diverse sensory profiles without overload/habituation, flattening/appropriating?

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